

 **Spaces**

 openadmet/

OpenADMET-ExpansionRx-Challenge 

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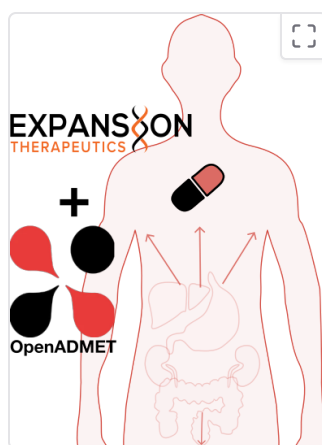
 

Welcome to the OpenADMET + ExpansionRx Blind Challenge!

Your task is to develop and submit predictive models for key ADMET properties on a blinded test set of real world drug discovery data 🧑🔬

Go to the **Leaderboard** to check out how the challenge is going. To participate, head out to the **Submit** tab and upload your results as a `CSV` file.

We released an intermediate leaderboard on December 2nd, with submissions evaluated against the full blinded test set. Check it out on the OpenADMET [blog](#)!

[About](#)[Leaderboard](#)[Submit](#)[FAQ](#)

OpenADMET + ExpansionRx

Computational Blind Challenge in ADMET

This challenge is a community-driven initiative to benchmark predictive models for ADMET properties in drug discovery, hosted by **OpenADMET** in collaboration with **ExpansionRx**.

Why are ADMET properties important in drug discovery?

Small molecules continue to be the bricks and mortar of drug discovery globally, accounting for ~75% of FDA approvals over the last decade. Oral bioavailability, easily tunable properties, modulation of a wide range of mechanisms, and ease of

manufacturing make small molecules highly attractive as therapeutic agents. Moreover, emerging small molecule modalities such as degraders, expression modulators, molecular glues, and antibody-drug conjugates (to name a few) have vastly expanded what we thought small molecules were capable of. It is fairly difficult to predict the lifetime and distribution of small molecules within the body. Additionally, interaction with off-targets can cause safety issues and toxicity. Collectively these *Absorption, Distribution, Metabolism, Excretion, Toxicology*--or **ADMET**--properties sit in the middle of the assay cascade and can make or break preclinical candidate molecules. **OpenADMET** aims to address these challenges through an open science effort to build predictive models of ADMET properties by characterizing the proteins and mechanisms that give rise to these properties through integrated structural biology, high throughput experimentation and integrative computational models. Read more about our strategy to transform drug discovery on our [website](#). Critical to our mission is developing open datasets and running community blind challenges to assess the current state of the art in ADMET modeling. Building on the success of the recent [ASAP-Polaris-OpenADMET blind challenge](#) in computational methods for drug discovery, we bring you a brand new challenge in collaboration with **ExpansionRx**. During a recent series of drug discovery campaigns for RNA mediated diseases, ExpansionRX collected a variety of ADMET data for off-targets and properties of interest, which they are generously sharing with the community for this challenge.

The Challenge

Participants will be tasked with solving real-world ADMET prediction problems ExpansionRx faced during lead optimization. Specifically, you will be asked to predict the ADMET properties of late-stage molecules based on earlier-stage data from the same campaigns. For this challenge we selected nine (9) crucial endpoints for the community to predict:

- LogD
- Kinetic Solubility **KSOL**: μM
- Mouse Liver Microsomal (**MLM**) *CLint*: $\text{mL}/\text{min}/\text{kg}$
- Human Liver Microsomal (**HLM**) *CLint*: $\text{mL}/\text{min}/\text{kg}$
- Caco-2 Efflux Ratio
- Caco-2 Papp A>B ($10^{-6} \text{ cm}/\text{s}$)
- Mouse Plasma Protein Binding (**MPPB**): % Unbound
- Mouse Brain Protein Binding (**MBPB**): % Unbound

- Mouse Gastrocnemius Muscle Binding (**MGMB**): % Unbound

Find more information about these endpoints on our [blog](#).

UPDATE: The Challenge is now live! Data available at the following Hugging Face Datasets

- Training: <https://huggingface.co/datasets/openadmet/openadmet-expansionrx-challenge-train-data>
- Test: <https://huggingface.co/datasets/openadmet/openadmet-expansionrx-challenge-test-data-blinded>

You can also watch a [Webinar](#) where we introduce the challenge, hosted by [Collaborative Drug Discovery \(CDD\)](#).

UPDATE: [OpenEye Cadence Molecular Sciences](#) is generously providing access to their Toolkit for interested participants during the duration of the challenge. Request access by filling out this [Google Form](#).

✓ How to Participate

1. **Register:** Create an account with Hugging Face.
2. **Walk through the tutorials:** We have prepared a [Tutorial](#) showing how to train a model and submit to the leaderboard.
3. **Download the Public Dataset:** Download the ExpansionRx [training](#) and [blinded test](#) sets from Hugging Face.
4. **Train Your Model:** Use the provided training data for each ADMET property of your choice.
5. **Submit Predictions:** Follow the instructions in the *Submit* tab to upload your predictions.
6. Join the discussion on the [Challenge Discord](#)!

Data:

The training set contains the following parameters:

Column	Unit	Type	Description
Molecule Name		str	Identifier for the molecule
Smiles		str	Text representation of the 2D molecular structure

LogD		float	LogD
KSol	uM	float	Kinetic Solubility
MLM CLint	mL/min/kg	float	Mouse Liver Microsomal
HLM CLint	mL/min/kg	float	Human Liver Microsomal
Caco-2 Permeability Efflux		float	Caco-2 Permeability Efflux Ratio
Caco-2 Permeability Papp A>B	10 ⁻⁶ cm/s	float	Caco-2 Permeability Papp A>B
MPPB	% Unbound	float	Mouse Plasma Protein Binding
MBPB	% Unbound	float	Mouse Brain Protein Binding
MGMB	% Unbound	float	Mouse Gastrocnemius Muscle Binding

You can download the training data from the [Hugging Face dataset](#).

The test set will remain blinded until the challenge submission deadline. You will be tasked with predicting the same set of ADMET endpoints for the test set molecules.

The training and blinded test set will also be made available on the [CDD Vault](#). An account to access the CDD Vault can be requested by filling out this [form](#). Note that by joining the Vault, your account will be visible to other participants, so this